NETWORK DESIGN & SECURITY PLAN

Prepared for: ***[CLIENT ORGANIZATION NAME]***

[Site Location / Address]

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How to Use This Document

This document is your network design plan. It describes how your organization's network is structured, what security protections are in place, and how each part of the system works together to keep your data and devices safe.

You do not need a technical background to read this document. Each section is written in plain language. Where technical terms are used, a brief explanation is provided in context.

Sections marked with [FILL IN] contain placeholders that should be completed with information specific to your organization. Your network engineer or IT provider will help you complete these fields.

Note: All placeholder fields appear in bold blue italics like this: [Example Field]. Replace each one with the appropriate information before finalizing this document.

1. Executive Summary

This section provides a plain-language overview of the network being designed or documented. It is intended for leadership, ownership, and non-technical stakeholders who need to understand the scope and purpose of the network without diving into technical details.

1.1 Purpose of This Plan

This network plan documents the design, layout, and security configuration of the network at:

Organization: ***[Client Organization Name]***
Location: ***[Site Address or Description]***
Scope: ***[Brief description — e.g., 'Single-site office network supporting 20 users and 3 servers']***

1.2 What This Plan Covers

This document covers:

- The physical equipment installed on the network (routers, switches, wireless access points)
- How the network is divided into separate zones to protect sensitive systems
- Which devices are assigned to which zones and what addresses they use
- What security rules are in place to control who can communicate with what
- Steps taken to harden the network against common threats

1.3 Why Network Segmentation Matters

Think of network segmentation like separating rooms in a building with locked doors. Even if someone gains access to one room (for example, a guest WiFi network), they cannot walk into a restricted area (such as your accounting systems or servers) without separate authorization.

This plan divides the network into distinct zones. Each zone has its own rules about what it can access and what can access it. This limits damage if any single device is ever compromised.

2. Hardware Inventory

The table below lists every physical device that makes up the network infrastructure. These are the hardware components — the physical boxes that all other services and devices depend on.

> **Plain Language:** The 'Management IP' column shows the private address used to configure and manage each device. These addresses are only accessible by authorized administrators and are never exposed to the internet.

Device	Model	Role / Function	Zone	Management Address
Firewall / Router	[object Object]	Controls all traffic in and out of the network. Acts	[object Object]	[object Object]

Device	Model	Role / Function	Zone	Management Address
		as the main security checkpoint.		
Managed Switch	[object Object]	Connects all wired devices and enforces traffic separation between zones.	[object Object]	[object Object]
Server / Host	[object Object]	Hosts virtual machines and internal services.	[object Object]	[object Object]
Wireless Access Point	[object Object]	Provides WiFi. Separates staff and guest networks into different zones.	[object Object]	[object Object]
ISP Modem / Gateway	ISP Provided	Connects the organization to the internet.	WAN (Internet)	ISP Assigned
[object Object]	[object Object]	[object Object]	[object Object]	[object Object]

Add or remove rows as needed to reflect all physical infrastructure at this site.

3. Network Zone Design (VLAN Layout)

The network is divided into separate zones called VLANs (Virtual Local Area Networks). Each VLAN is like a separate, isolated network within the same physical infrastructure. Devices in one zone cannot communicate with devices in another zone unless a specific rule permits it.

Why This Matters: If a device on your guest WiFi is infected with malware, it cannot reach your internal servers or workstations because they are in separate zones with no connection between them.

Zone #	Zone Name	Purpose (Plain Language)	Address Range	Gateway	Notes
[object Object]	Staff / Users	Personal computers, phones, and tablets used by employees.	[object Object]	[object Object]	Staff WiFi maps here
[object Object]	Servers / Internal	Internal servers, databases, and hosted applications.	[object Object]	[object Object]	Restricted access
[object Object]	IoT / Devices	Smart devices, printers, cameras, or other equipment that	[object Object]	[object Object]	Isolated from staff

Zone #	Zone Name	Purpose (Plain Language)	Address Range	Gateway	Notes
		connects to the network.			
[object Object]	Public-Facing	Services accessible from the internet (website, client portal, etc.).	[object Object]	[object Object]	Strictly controlled
[object Object]	Guest WiFi	Visitors and contractors. Internet access only, no internal access.	[object Object]	[object Object]	No internal access
[object Object]	Management	Administrator access to configure routers, switches, and servers. No regular users.	[object Object]	[object Object]	Static IPs only
[object Object]	Unused / Disabled	Any unused network port is placed here so it has no network access.	N/A	N/A	Security default

Customize zone names and numbers to match your organization's structure. Each zone should reflect a distinct group of users or device types.

4. Device Address Assignments

Every device on the network is assigned an IP address — a unique identifier, similar to a street address, that allows devices to find and communicate with each other. Infrastructure devices (routers, switches, servers) use fixed addresses that never change, called static IPs. End-user devices (laptops, phones) typically receive addresses automatically from a range.

4.1 Management Zone — Infrastructure Devices

Device	IP Address	Access Method	Notes
[object Object]	[object Object]	Secure web browser (HTTPS)	Admin access restricted to Management Zone only
[object Object]	[object Object]	SSH (encrypted command line)	Password or key-based login required
[object Object]	[object Object]	Secure web browser (HTTPS)	Management only — client traffic is separate
[object Object]	[object Object]	[object Object]	[object Object]

4.2 Server Zone — Hosted Services

Service / VM	IP Address	Port(s)	Notes
[object Object]	[object Object]	[object Object]	[object Object]
[object Object]	[object Object]	[object Object]	[object Object]
[object Object]	[object Object]	[object Object]	[object Object]
[object Object]	[object Object]	[object Object]	[object Object]

4.3 Public-Facing Zone — Internet-Accessible Services

These services can be reached from the internet but are strictly controlled. All external traffic passes through a reverse proxy (a security intermediary) before reaching any internal service.

Service	IP Address	Public Port(s)	Notes
[object Object]	[object Object]	80, 443	All web traffic enters here first; acts as a security gatekeeper
[object Object]	[object Object]	[object Object]	Routed through web gateway; not directly reachable from internet
[object Object]	[object Object]	[object Object]	[object Object]

5. Switch Port Configuration

A managed switch is the physical device that all wired network cables connect to. It controls which zone each port belongs to and enforces separation between zones at the hardware level.

> **Plain Language:** Every cable jack on the wall connects to a port on this switch. Ports that are not in active use are disabled so that someone cannot plug in an unauthorized device and gain network access.

Port	Connected Device	Type	Zone(s)	Notes
1	[object Object]	Trunk (carries all zones)	All	Main connection between firewall and switch
2	[object Object]	Trunk (carries server zones)	[object Object]	Carries all virtual machine traffic
3	[object Object]	Trunk (staff + IoT zones)	[object Object]	Staff and IoT WiFi separated at AP level
4	DISABLED — Future Use	Shutdown	999 (no access)	Re-enable only when a specific device is ready to connect

Port	Connected Device	Type	Zone(s)	Notes
5	DISABLED — Unused	Shutdown	999 (no access)	Disabled until needed
6	DISABLED — Unused	Shutdown	999 (no access)	Disabled until needed
7	DISABLED — Unused	Shutdown	999 (no access)	Disabled until needed
8	[object Object]	Access (Management only)	99	Used only for authorized administrative access

5.1 Port Security Controls Applied

In addition to disabling unused ports, the following protections are configured on the switch:

Security Control	What It Does (Plain Language)
Port Shutdown	Unused ports are administratively turned off. An unplugged, disabled port cannot be used to access the network.
DHCP Snooping	Prevents a rogue or unauthorized device from acting as a network address server and redirecting other devices' traffic through itself.
ARP Inspection	Prevents a technique called 'ARP poisoning' where an attacker tricks devices into sending their data to the wrong destination — a common way to intercept communications.
MAC Address Locking	Certain ports (such as camera or IoT ports) are locked to a specific device's hardware ID. If any other device is plugged in, the port blocks it automatically.
Discovery Protocol Disabled	The switch does not broadcast its make, model, or configuration to connected devices. This prevents attackers from easily profiling the network.
Encrypted Remote Access Only	All remote management of the switch requires an encrypted connection (SSH). Unencrypted connections (Telnet) are blocked entirely.

6. Firewall & Access Control Rules

The firewall acts as a security checkpoint that inspects every request to communicate between zones or with the internet. Traffic is blocked by default — only connections that are explicitly permitted are allowed through.

Default Deny Policy: If a connection is not specifically listed as allowed, the firewall automatically blocks it. This means the organization's exposure is limited to only what has been intentionally opened.

6.1 Zone-to-Zone Access Rules

From Zone	To Destination	Decision	Reason
Staff / Users	Internet	ALLOW	Employees need unrestricted internet access for work
Staff / Users	Internal Servers	ALLOW	Staff need access to internal applications and file servers
Staff / Users	Management Zone	BLOCK	Regular users should never be able to access network configuration interfaces
IoT / Devices	Internet	BLOCK	Smart devices and cameras do not need internet access in most configurations
IoT / Devices	Other Zones	BLOCK	Isolating IoT devices prevents compromised equipment from reaching sensitive systems
Public-Facing	Internal Network	BLOCK	A public-facing server should never be able to reach internal systems — limits breach impact
Guest WiFi	Internet	ALLOW	Guests need internet access only
Guest WiFi	Internal Zones	BLOCK	Guest traffic must never reach staff systems or servers
Management	All Infrastructure	ALLOW	Admins need access to configure all infrastructure devices
Management	Internet	BLOCK	Management tools should never have outbound internet access — reduces attack surface

6.2 Internet-Inbound Rules (What Can Reach You From Outside)

By default, nothing from the internet can reach your internal network. The only exceptions are the specific services listed below.

Port / Protocol	Destination Service	Purpose
80, 443 (Web)	[object Object]	Standard web traffic. All requests pass through the web gateway first.
[object Object]	[object Object]	[object Object]
51820 (VPN)	Remote Admin VPN	Encrypted tunnel for authorized remote administration only. No other remote access method is permitted.

6.3 Blocked Dangerous Ports

The following services are explicitly blocked at the firewall on all zones regardless of whether they are running. These represent high-risk services with histories of exploitation.

Port(s)	Service Name	Why It Is Blocked
20, 21	FTP (File Transfer)	Transmits files and passwords in plain text with no encryption. Replaced by secure alternatives.

Port(s)	Service Name	Why It Is Blocked
23	Telnet	Sends all data including passwords in plain text. Fully replaced by SSH.
25	SMTP (Email Relay)	Should never be open on a business network — actively scanned by spammers to send bulk email.
139, 445	Windows File Sharing	Historically exploited by major attacks including WannaCry. File sharing handled through secure methods only.
3389	Remote Desktop (RDP)	Heavily targeted on the internet. Remote access is handled through VPN only.
5900+	VNC Remote Desktop	Weak authentication history. Blocked in favor of VPN-based remote access.
3306	Database (MySQL)	Database services should only be accessible locally within their own server — never across the network.

7. Wireless Network Configuration

The wireless access point broadcasts two or more separate networks. Each WiFi network is mapped to a different zone, so a device connected to the guest network is completely separated from devices on the staff network at the hardware level.

Network Name (SSID)	Zone	Security Protocol	Who Should Use It
[object Object]	[object Object]	WPA3 / WPA2	Employees and authorized devices only
[object Object]	[object Object]	WPA2	Visitors, contractors, personal devices — internet only
[object Object]	[object Object]	WPA2	Smart TVs, printers, cameras, or other connected devices

Important: The access point's own configuration interface is only accessible from the Management Zone. It cannot be reached from any WiFi network, including staff WiFi.

8. Hosted Services & Applications

The following services are hosted internally on the organization's server infrastructure. Each service is isolated to a specific zone and configured to only expose the access it requires.

Service / Application	Zone	IP Address	RAM	Purpose
[object Object]	[object Object]	[object Object]	[object Object]	Internal file sharing and document management
[object Object]	[object Object]	[object Object]	[object Object]	Security front-end for all internet-accessible services
[object Object]	[object Object]	[object Object]	[object Object]	[object Object]
[object Object]	[object Object]	[object Object]	[object Object]	Filters malicious and unwanted network traffic at the DNS level
[object Object]	[object Object]	[object Object]	[object Object]	[object Object]

8.1 Resource Summary

Use this table to track total server resource consumption and plan capacity.

Component	Estimated RAM Usage
Host Server Overhead	~2 GB
[object Object]	[object Object]
[object Object]	[object Object]
[object Object]	[object Object]
TOTAL	**[X GB of Y GB available]**

9. Remote Access & VPN

Remote access to internal systems is only permitted through an encrypted VPN tunnel. No other method of remote access (such as Remote Desktop or direct port access) is enabled. This ensures that anyone accessing the network remotely is authenticated and their connection is encrypted end-to-end.

Setting	Configuration
VPN Type	[object Object]
Port / Protocol	[object Object]
Who Can Use It	[object Object]
Authentication Method	[object Object]
What Remote Users Can Access	[object Object]
Other Remote Methods	NONE — all other remote access methods are blocked

10. Recommended Deployment Order

Follow this sequence when building or deploying the network. Each step should be verified before moving to the next. This order ensures security controls are in place before any user devices or services are connected.

Step	Task	What Happens & Why
1	Configure Switch	Create all network zones, disable all unused ports, and apply port security before connecting any devices. This is the most critical first step.
2	Configure Firewall Zones	Set up all zone interfaces on the firewall with correct address ranges. Immediately disable the firewall's admin interface on all zones except the Management Zone.
3	Build Firewall Rules	Apply inter-zone blocking rules first, then add specific permit rules. Block all dangerous ports listed in Section 6.
4	Deploy DNS / Filtering Service	Spin up DNS filtering service. Update firewall DHCP settings to direct all DNS queries through it. Force all devices to use it regardless of their own settings.
5	Configure Wireless Access Point	Connect AP to switch. Create staff and guest WiFi networks and confirm each maps to the correct zone. Confirm management interface is only accessible from Management Zone.
6	Deploy Internal Services	Bring up file servers, internal applications, and business tools. Assign static addresses. Confirm zone isolation rules are in place before going live.
7	Deploy Public-Facing Services	Set up web gateway and any public services. Verify that public-facing servers cannot initiate connections to internal zones before exposing them to the internet.
8	Configure VPN	Set up and test VPN remote access. Confirm authorized users can reach designated internal resources. Test from an external connection before finalizing.
9	Deploy IoT / Camera Zone	Only when physical devices are in hand. Re-enable the port, configure the device, apply MAC locking. Never open a port in advance.

11. Security Hardening Checklist

Use this checklist to track completion of security controls during deployment. Each item represents a specific security action that must be verified before the network is considered production-ready.

Switch Hardening

- Disable all unused switch ports and assign them to the dead zone (no network access)
- Enable DHCP Snooping — only the firewall uplink port is trusted
- Enable ARP Inspection — only the firewall uplink port is trusted
- Disable device discovery protocols (CDP and LLDP) globally
- Block Telnet on all lines — enforce SSH version 2 only
- Apply MAC address locking on IoT and camera ports when deployed

Firewall Hardening

- Block high-risk ports (FTP, Telnet, SMTP, SMB, RDP, VNC) on all zone interfaces
- Configure default deny rules — only explicitly permitted traffic passes
- Disable firewall admin interface on all zones except Management
- Enable SSH with key-based authentication — disable password login
- Block Public-Facing Zone from initiating connections to internal zones
- Force all DNS traffic through internal DNS filtering service regardless of device settings
- VPN is the only approved method for remote administrative access

Server & Service Hardening

- Check open ports on every server after deployment — close any not on the approved list
- Stop and disable any unneeded services — do not just block the port
- Enable host-based firewall (UFW) on all Linux servers with default deny incoming
- Disable SSH password authentication on all servers — use key-based auth only
- Confirm Proxmox / hypervisor management interface accessible from Management Zone and Staff Zone only

Wireless Hardening

- Staff WiFi uses WPA3 / WPA2 with a strong passphrase
- Guest WiFi is isolated — internet only, no internal access
- AP management interface is only reachable from Management Zone
- IoT WiFi is isolated from staff network

Future / Recommended Enhancements

- Install IDS/IPS (intrusion detection/prevention) on the firewall WAN interface
- Configure DNS-over-HTTPS to encrypt DNS queries to upstream resolvers
- Apply protected ports on the switch to prevent lateral scanning between devices on the same zone
- Disable ICMP (ping) on the WAN interface to reduce internet-facing visibility
- Schedule quarterly review of firewall rules and open ports

12. Approval & Sign-Off

This section documents that the network plan has been reviewed, accepted, and approved by the appropriate stakeholders prior to implementation.

Role	Name	Date	Signature
Network Engineer / Designer	[object Object]	[object Object]	
Client / Organization Lead	[object Object]	[object Object]	
IT Manager / CTO	[object Object]	[object Object]	
[object Object]	[object Object]	[object Object]	

Document Revision History

Version	Date	Author	Summary of Changes
[object Object]	[object Object]	[object Object]	Initial draft
[object Object]	[object Object]	[object Object]	[object Object]
[object Object]	[object Object]	[object Object]	[object Object]

Appendix A: Glossary of Terms

This glossary defines technical terms used in this document in plain language.

Term	Plain Language Definition
VLAN	A virtual network zone. Divides one physical network into multiple isolated segments, like separate rooms in a building.
Firewall	A device that inspects all network traffic and blocks anything that violates security rules.
IP Address	A unique number assigned to each device on a network, similar to a mailing address.

Term	Plain Language Definition
Static IP	An IP address that never changes. Used for infrastructure devices so they are always reachable at the same address.
DHCP	A service that automatically assigns IP addresses to devices when they connect to the network.
DNS	Translates website names (like google.com) into IP addresses that computers can use.
Port	A numbered channel a service listens on. Think of it like an apartment number — the IP address is the building, the port is the unit.
VPN	A Virtual Private Network. Creates an encrypted tunnel so remote users can securely access the internal network over the internet.
SSH	Secure Shell. An encrypted method of remotely accessing and managing servers and network devices via command line.
Reverse Proxy	A security gateway that sits in front of web services. All external requests pass through it first before reaching the actual server.
DHCP Snooping	A switch-level protection that blocks unauthorized devices from acting as a network address server.
ARP Poisoning	An attack where a bad actor tricks devices into sending their data to the attacker's machine instead of the intended destination.
Default Deny	A firewall policy where all traffic is blocked unless a specific rule explicitly permits it.
MAC Address	A unique hardware identifier built into every network device. Used to identify specific devices regardless of their IP address.
IDS / IPS	Intrusion Detection / Prevention System. Monitors network traffic for suspicious patterns and can block attacks automatically.
WAN	Wide Area Network — the internet, or the connection between the organization's network and the outside world.

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